

Deep Networks

Gradient Descent and Optimization

Introduction to Deep Learning · Class 13

Outline

The Training Objective

Descent on a Convex Loss

Stochastic and Mini-batch Gradient Descent

Putting It Together

The Training Objective

From the gradient to the step

Class 11 ended with a gradient in hand. This class asks what to do with it.

Class	The question	The object
11	How do we obtain ∇E at all?	the algorithm
13	What is the step, and does it go down?	the direction and its estimate
14	How fast, and under what conditions?	the dynamics
15	How do we keep the signals well-scaled?	the conditioning

Today's plan: start with a loss on which descent *cannot* fail, understand exactly why, and then remove that assumption one piece at a time.

Notation

$\mathbf{x}, y, \hat{y}$	input, target, prediction	η	learning rate
$\mathbf{w}$	every learnable parameter	τ	iteration index
$E(\mathbf{w})$	the error function being minimized	N	training-set size
$E_n(\mathbf{w})$	its contribution from point n	B	mini-batch size
∇E	gradient of E with respect to $\mathbf{w}$	W	number of parameters
$\mathbf{H}$	Hessian, $\nabla \nabla E$	$\lambda_i, \mathbf{u}_i$	eigenvalues, eigenvectors of $\mathbf{H}$

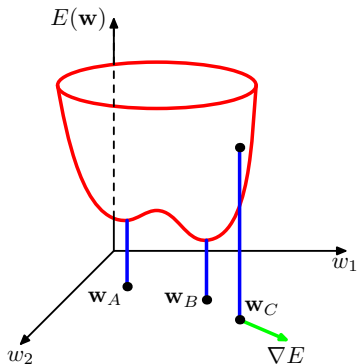
The training objective

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} E(\mathbf{w}), \quad E(\mathbf{w}) = \sum_{n=1}^N E_n(\mathbf{w})$$

- Class 6 built $E(\mathbf{w})$ from maximum likelihood; Class 11 made ∇E computable
- E is a **sum over data points** because the likelihood of independent observations is a product
- No closed form solves $\nabla E(\mathbf{w}) = 0$ for a network, so we iterate

Optimization is a means, not the goal: we want good predictions on *new* data, so we will happily stop short of the exact minimum.

The error surface



E viewed as a surface over weight space. $\mathbf{w}_A$ is a local minimum and $\mathbf{w}_B$ the global minimum; at $\mathbf{w}_C$ the local slope is ∇E . Bishop & Bishop (2024), Fig. 7.1.

The hiker in fog

On the mountain

the altitude at your feet

the slope you can feel underfoot

fog: no view of the far side

the length of one stride

the valley you happen to start in

In training

$E(\mathbf{w})$ at the current $\mathbf{w}$

$\nabla E(\mathbf{w})$

no access to the global shape of E

the learning rate η

the initialization

- The hiker knows the slope **exactly** and the map **not at all**
- Every design choice today is about how far to trust one local slope reading

Descent on a Convex Loss

Convexity

Definition (convex function)

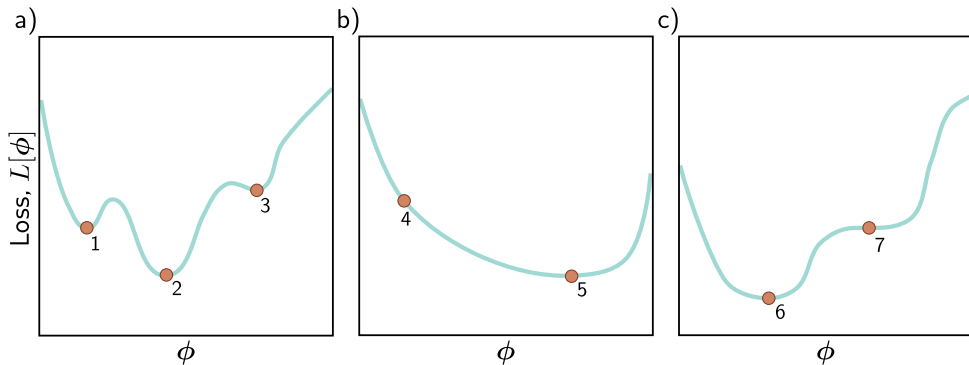
E is **convex** if no chord lies below its graph:

$$E(\theta \mathbf{u} + (1 - \theta)\mathbf{v}) \leq \theta E(\mathbf{u}) + (1 - \theta)E(\mathbf{v}) \quad \forall \mathbf{u}, \mathbf{v}, \theta \in [0, 1].$$

- Where E is twice differentiable this says the Hessian $\mathbf{H}$ is positive semi-definite **everywhere**
- Least squares for a **linear** model is convex
- So is logistic regression, and every generalised linear model of Class 6

Prince (2023), §6.1.2.

Convex and non-convex, side by side



The chord test. On the left every chord lies above the graph; on the right a single chord cutting below is enough to disqualify the function. Prince (2023), §6.1.2.

What convexity buys

- Every stationary point is a **global** minimum — there is nowhere bad to get stuck
- The set of minimisers is connected: no isolated wrong answers
- Wherever we start, walking downhill reaches the best solution
- Progress can be measured: $E(\mathbf{w}) - E(\hat{\mathbf{w}})$ is a meaningful number that decreases

On a convex loss the training procedure *cannot* fail. Sections 2 and 3 develop gradient descent under this assumption; Section 4 removes it and counts the cost.

Gradient descent

Every method in this class and the next has the same shape,

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} + \Delta \mathbf{w}^{(\tau-1)},$$

and differs only in the choice of $\Delta \mathbf{w}$. Today,

Definition (batch gradient descent)

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \eta \nabla E(\mathbf{w}^{(\tau-1)}), \quad \eta > 0.$$

Each ∇E uses the whole training set, which is what makes the method a **batch** method.

Bishop & Bishop (2024), Eqs. 7.15–7.16; Prince (2023), Eqs. 6.2–6.3.

Why the step goes along $-\nabla E$

A small step changes the error, to first order, by $\Delta E \simeq \Delta \mathbf{w}^\top \nabla E(\mathbf{w})$.

Lemma 1 (The negative gradient is the steepest direction)

Fix $\mathbf{w}$ with $\nabla E(\mathbf{w}) \neq 0$. Among all unit vectors $\mathbf{d}$, the quantity $\mathbf{d}^\top \nabla E(\mathbf{w})$ is minimised uniquely at $\mathbf{d}^ = -\nabla E(\mathbf{w}) / \|\nabla E(\mathbf{w})\|$, where it equals $-\|\nabla E(\mathbf{w})\|$.*

- **By Cauchy-Schwarz, $|\mathbf{d}^\top \nabla E| \leq \|\mathbf{d}\| \|\nabla E\| = \|\nabla E\|$ for any unit $\mathbf{d}$.**
- So $\mathbf{d}^\top \nabla E \geq -\|\nabla E\|$: no unit direction beats $-\|\nabla E\|$.
- Equality holds only when $\mathbf{d}$ is parallel to ∇E ; the sign forces *anti*-parallel.
- So $\mathbf{d}^*$ attains the bound and is the unique minimiser. □

Bishop & Bishop (2024), Eqs. 7.1–7.2.

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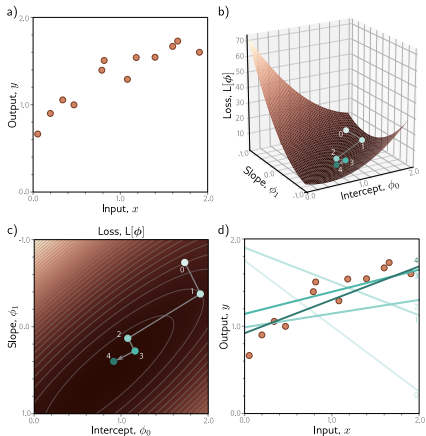
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Bishop & Bishop (2024), Eqs. 7.1–7.2.

Descent on a convex loss



- (a) the training set; (b) the loss surface, four iterations marked
- (c) the same iterations as a heat map
- (d) the fitted line at each, lightest first
- Least squares for a **linear** model is convex, so this cannot fail

Prince (2023), Fig. 6.1. His ϕ_0, ϕ_1 are our w_0, w_1 .

Two ways to choose the step length

A fixed learning rate

- η chosen once, before training
- One gradient evaluation per step
- The only affordable option once W is large

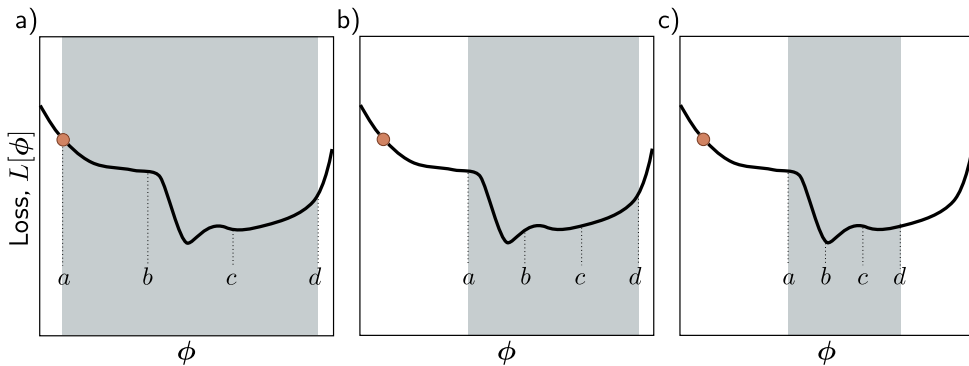
A line search

- Try several η and keep the best
- Many evaluations per step
- Extracts the most progress available along this direction

A line search is exact and unaffordable; a fixed η is crude and cheap. Deep learning takes the cheap one and spends the saving on many more steps.

Prince (2023), §6.1.

Choosing the step length by search



A line search evaluates several step lengths along the descent direction and keeps the one that lowers the loss most. Prince (2023), §6.1.

Four regimes for the step size

Stride	Condition	What happens
tiny	$\eta \ll 1/L$	every step helps, but barely: the hiker crawls
right	$\eta = 1/L$	the guaranteed decrease is largest
long	$1/L < \eta < 2/L$	still downhill, and it starts to oscillate
too long	$\eta > 2/L$	the step overshoots the bend and E can increase

- Nobody knows L for a deep network, so η is tuned by experiment
- A loss that diverges is the symptom of having crossed $2/L$, which is why lowering η is the first thing to try

Why gradients are worth their cost

The surface near a minimum is described by $\mathbf{b}$ and $\mathbf{H}$, so locating it means pinning those down.

$$\underbrace{W}_{\text{entries of } \mathbf{b}} + \underbrace{\frac{W(W+1)}{2}}_{\text{independent entries of the symmetric } \mathbf{H}} = \frac{W(W+3)}{2} = O(W^2) \text{ numbers}$$

- **Function values only:** one number per evaluation, so $O(W^2)$ evaluations at $O(W)$ work each $\Rightarrow O(W^3)$
- **Gradients:** W numbers per evaluation, so $O(W)$ evaluations, each $O(W)$ by backpropagation $\Rightarrow O(W^2)$

A factor of W — for a million parameters, a factor of a million. B&B (2024), §7.2.1.

The local quadratic model

Convexity tells us where we are going. Curvature tells us how large a step is safe.

$$E(\mathbf{w}) \simeq E(\hat{\mathbf{w}}) + (\mathbf{w} - \hat{\mathbf{w}})^\top \mathbf{b} + \frac{1}{2}(\mathbf{w} - \hat{\mathbf{w}})^\top \mathbf{H}(\mathbf{w} - \hat{\mathbf{w}}),$$

$$\mathbf{b} \equiv \nabla E|_{\mathbf{w}=\hat{\mathbf{w}}}, \quad \mathbf{H}(\hat{\mathbf{w}}) \equiv \nabla \nabla E(\mathbf{w})|_{\mathbf{w}=\hat{\mathbf{w}}}.$$

- $\mathbf{b}$ has length W ; $\mathbf{H}$ is $W \times W$ and symmetric
- At a minimum $\mathbf{w}^*$ the linear term vanishes, leaving
$$E(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2}(\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}(\mathbf{w} - \mathbf{w}^*)$$
- Everything in this section is read off that one expression

Bishop & Bishop (2024), Eqs. 7.3–7.7.

Rotating into the eigenvectors of H

With $H\mathbf{u}_i = \lambda_i\mathbf{u}_i$ and $\mathbf{u}_i^T\mathbf{u}_j = \delta_{ij}$, expand the displacement as $\mathbf{w} - \mathbf{w}^* = \sum_i \alpha_i \mathbf{u}_i$.
Then

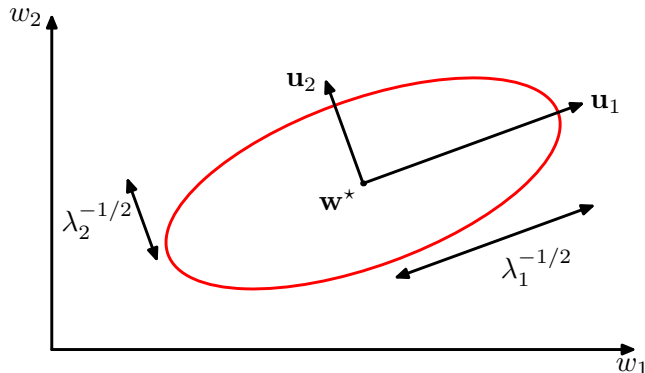
$$E(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2} \sum_i \lambda_i \alpha_i^2.$$

- A change of coordinates: origin at $\mathbf{w}^*$, axes along the eigenvectors
- Every cross term cancels, by orthonormality

In these coordinates E is a *sum of independent one-dimensional parabolas*, one per eigenvector, the i th with curvature λ_i . Nothing couples them.

Bishop & Bishop (2024), Eqs. 7.8–7.11.

Reading the curvature off the contours



Contours of the quadratic model are ellipses centred on w^* , with axes along the eigenvectors u_i of H and semi-axis lengths proportional to $\lambda_i^{-1/2}$. Bishop & Bishop (2024), Fig. 7.2.

Smoothness: one number for the worst curvature

Definition (L -smooth)

E is L -smooth if its gradient is Lipschitz,

$$\|\nabla E(\mathbf{u}) - \nabla E(\mathbf{v})\| \leq L\|\mathbf{u} - \mathbf{v}\| \quad \forall \mathbf{u}, \mathbf{v}.$$

- Equivalently, wherever E is twice differentiable, every eigenvalue satisfies $|\lambda_i| \leq L$: it is the **sharpest bend anywhere** in the landscape
- For a quadratic the curvature is the same everywhere, so $L = \lambda_{\max}$ — the curvature of the sharpest cross-section, and the *shortest* axis of the ellipse
- L is a property of the **problem**: we do not choose it, and in a network we do not know it

How far a single step may go

Lemma 2 (One step is guaranteed to go downhill)

Let E be L -smooth and let $\mathbf{w}^+ = \mathbf{w} - \eta \nabla E(\mathbf{w})$. Then

$$E(\mathbf{w}^+) \leq E(\mathbf{w}) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla E(\mathbf{w})\|^2.$$

In particular $E(\mathbf{w}^+) < E(\mathbf{w})$ whenever $0 < \eta < 2/L$ and $\nabla E(\mathbf{w}) \neq 0$; the bound is tightest at $\eta = 1/L$.

- L -smoothness gives $E(\mathbf{v}) \leq E(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^\top \nabla E(\mathbf{w}) + \frac{L}{2} \|\mathbf{v} - \mathbf{w}\|^2$.
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- Collecting the $\|\nabla E\|^2$ terms gives the bound; $\eta(1 - L\eta/2) > 0$ on $(0, 2/L)$ and is maximal at $\eta = 1/L$. □

Nesterov (2004), Lemma 1.2.3.

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Removing the assumption

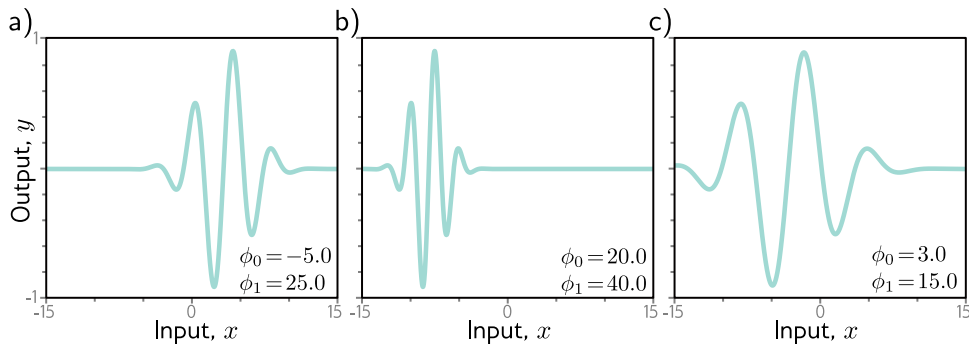
Everything so far assumed a convex loss. Shallow and deep networks are not convex.

- The steepest-descent lemma survives: it is local, and uses only ∇E
- The descent lemma survives: it is local, and uses only smoothness
- What does **not** survive is the link between “stationary” and “best”

Each step is still guaranteed to lower E . Nothing guarantees any more that the point it converges to is the one we wanted.

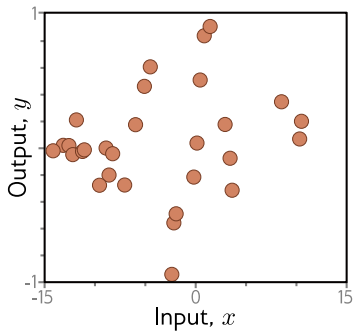
Prince (2023), §6.1.2.

A two-parameter model we can draw



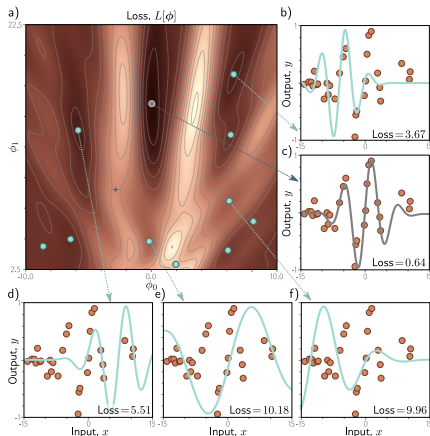
A model with two parameters: w_0 moves it, w_1 squeezes it. Prince (2023), Fig. 6.2, where these are ϕ_0 and ϕ_1 .

Its training set



28 points, made by sampling the model at $\mathbf{w} = [0.0, 16.6]^T$ and adding Gaussian noise. Prince (2023), Fig. 6.3.

A loss that is not convex



- (a) many local minima (cyan), one global minimum (grey)
- The blue cross is a saddle point: $\nabla E = 0$, but the surface falls away in one direction
- (b)–(f) are the models at each minimum; none improves under a small change

Prince (2023), Fig. 6.4.

Stationary points

Proposition 3 (Classification of stationary points)

Let $\nabla E(\mathbf{w}^) = 0$ and let $\mathbf{H}$ be the Hessian at $\mathbf{w}^*$, with eigenvalues $\{\lambda_i\}$. Then $\mathbf{w}^*$ is a local minimum if and only if $\lambda_i > 0$ for all i ; a local maximum if and only if $\lambda_i < 0$ for all i ; and a saddle point if the signs are mixed.*

- **Fix j , set $\alpha_i = 0$ for $i \neq j$ and vary α_j : then $E - E(\mathbf{w}^*) = \frac{1}{2} \lambda_j \alpha_j^2$.**
- So each eigen-direction votes independently: E rises along $\mathbf{u}_j$ if $\lambda_j > 0$ and falls if $\lambda_j < 0$.
- Writing $\mathbf{v} = \sum_i c_i \mathbf{u}_i$ gives $\mathbf{v}^T \mathbf{H} \mathbf{v} = \sum_i c_i^2 \lambda_i$, so $\mathbf{H}$ is positive definite exactly when every $\lambda_i > 0$.
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Bishop & Bishop (2024), §7.1.1.

Saddle points are the practical problem

- At a saddle the gradient vanishes, but the surface falls away in at least one direction
- Gradient descent can leave a saddle — provided it is not sitting exactly on it
- But the surface nearby is **flat**, so $\|\nabla E\|$ is small over a large region
- Stopping when $\|\nabla E\|$ is small therefore risks stopping at a saddle and calling it convergence
- In high dimension saddles vastly outnumber local minima: a random sign pattern is far more likely to be mixed than all positive

Prince (2023), §6.1.3.

How many minima are there?

- A two-layer network with M hidden units has $M! 2^M$ parameter vectors computing the **same** function
- So every minimum arrives with a combinatorial family of copies — for $M = 100$, more than 10^{187} of them
- Beyond those there are **non-equivalent** minima: one global, many local. It was long feared these would trap gradient methods; in practice large networks reach comparable solutions from many different starts

Counting minima is the wrong worry. Their *quality* is the right one.

Bishop & Bishop (2024), §6.2.4, §7.1.

The bill for losing convexity

On a convex loss

every stationary point is global

the destination is the best solution

small $\|\nabla E\|$ means nearly done

one run suffices

On a network's loss

stationary says nothing about quality

the destination is decided by the start

it may mean a saddle

several runs, compared on held-out data

Deterministic descent inherits every flaw of the point it was dropped at. The remedy in the next section is not a better direction — it is a *noisier* one.

Stochastic and Mini-batch Gradient Descent

The cost of a batch step

- $E(\mathbf{w}) = \sum_{n=1}^N E_n(\mathbf{w})$, so one gradient costs N backward passes
- With N in the millions, a single parameter update becomes a major computation
- It is also **wasteful under redundancy**: duplicate every data point and the cost doubles while the direction is unchanged, up to a rescaling of η

We do not need the exact gradient. We need a cheap direction that is right *on average*.

Bishop & Bishop (2024), §7.2.3.

Stochastic gradient descent

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \eta \nabla E_n(\mathbf{w}^{(\tau-1)})$$

- One data point per update; a full pass over the data is an **epoch**
- Also called *online* gradient descent when the data arrives as a stream

Algorithm 4 (Stochastic gradient descent)

Input: data indexed by $n \in \{1, \dots, N\}$; per-point error $E_n(\mathbf{w})$; learning rate η ; initial $\mathbf{w}$
 $n \leftarrow 1$; **repeat** $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla E_n(\mathbf{w})$; $n \leftarrow n + 1 \pmod{N}$
until convergence; **return** $\mathbf{w}$

Bishop & Bishop (2024), Eq. 7.18, Alg. 7.1; Bottou (2010).

The estimate is unbiased

Proposition 5 (Mean and variance of the mini-batch gradient)

Let $\mathcal{B}$ be B indices drawn uniformly at random and let $\hat{\mathbf{g}} = \frac{N}{B} \sum_{n \in \mathcal{B}} \nabla E_n(\mathbf{w})$. Then $\mathbb{E}[\hat{\mathbf{g}}] = \nabla E(\mathbf{w})$, and each component of $\hat{\mathbf{g}}$ has standard deviation $N\sigma/\sqrt{B}$, where σ is the standard deviation of that component across the data.

- Each index is uniform on $\{1, \dots, N\}$, so $\mathbb{E}[\nabla E_n] = \frac{1}{N} \nabla E$.
- Linearity over the B draws gives $\mathbb{E}[\hat{\mathbf{g}}] = \frac{N}{B} B \frac{1}{N} \nabla E = \nabla E$: unbiased, for every B .
- $\hat{\mathbf{g}}$ is N times a sample mean of B draws, so its variance is $N^2 \sigma^2 / B$.
- The $\sqrt{B}$ in the denominator is the whole argument for mini-batches. □

Bishop & Bishop (2024), §7.2.4; Prince (2023), §6.2.1.

The estimate is unbiased

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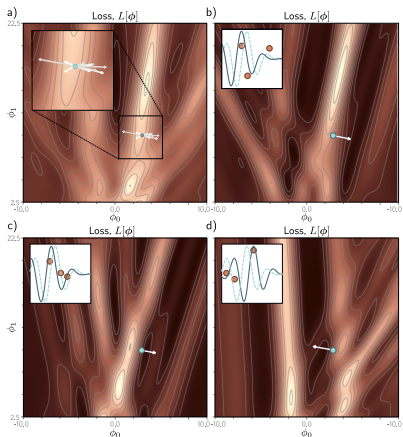
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- **The $\sqrt{B}$ in the denominator is the whole argument for mini-batches.** □

Bishop & Bishop (2024), §7.2.4; Prince (2023), §6.2.1.

Each batch is its own surface



- (a) the full loss, with the distribution of possible updates inset
- (b), (c) two batches, each with its own surface and its own downhill direction
- (d) a step that goes downhill on the batch but uphill on the full loss

Prince (2023), Fig. 6.6.

Mini-batches

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \eta \sum_{n \in \mathcal{B}_\tau} \nabla E_n(\mathbf{w}^{(\tau-1)})$$

- $B = 1$ is stochastic gradient descent, $B = N$ is batch descent, and everything useful lies in between
- Raising B a hundredfold improves the estimate tenfold and costs a hundredfold — **diminishing returns**
- Powers of two are common because they suit the hardware

The name does not change: it is still called “stochastic gradient descent” when $B > 1$. Prince (2023), Eq. 6.10; B&B (2024), §7.2.4.

Mini-batch stochastic gradient descent

Algorithm 6 (Mini-batch stochastic gradient descent)

Input: data indexed by $n \in \{1, \dots, N\}$; batch size B ; per-batch error $E_{n:n+B-1}(\mathbf{w})$; learning rate η ; initial $\mathbf{w}$

$n \leftarrow 1$

repeat

$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla E_{n:n+B-1}(\mathbf{w})$ *// weight vector update*

$n \leftarrow n + B$

if $n > N$ *then* shuffle data; $n \leftarrow 1$ *end if*

until convergence

return $\mathbf{w}$

Bishop & Bishop (2024), Algorithm 7.2.

What the batch size buys

- A small batch does not converge to a point; it settles into a **noise ball** whose radius grows as B shrinks
- Per *iteration* the full batch wins — it has the exact direction
- Per *gradient evaluation*, which is what costs money, a small batch is far ahead early
- The crossover is why practice uses a mini-batch, then raises B or lowers η late in training

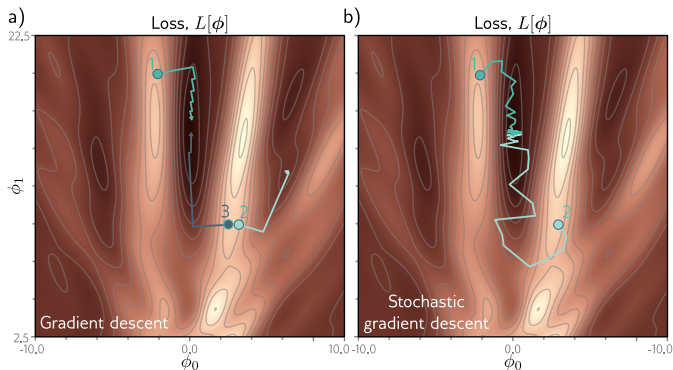
Choosing B is choosing where on this trade-off to sit, which is exactly why B is a hyperparameter and not a derived quantity.

Shuffling

- Raw data sets carry an order: collected by date, sorted alphabetically, grouped by class
- Consecutive blocks then make every batch unrepresentative, and Proposition 5 no longer applies — the draw is not uniform
- Shuffle the whole set, then take batches as successive blocks
- Reshuffling between epochs means a batch is unlikely to repeat, which also helps leave a poor valley

Unbiasedness was assumed, not granted. Shuffling is what makes the assumption true.

Noise is the feature



(a) gradient descent, whose destination is fixed by the start; (b) stochastic gradient descent, which can leave the wrong valley. Prince (2023), Fig. 6.5.

Six properties of stochastic gradient descent

1. It still improves the fit **to some subset** at every step, so the updates are sensible even when they are not optimal
2. Drawing without replacement means every example contributes equally over an epoch
3. It is cheaper per step by a factor of N/B
4. It can **escape local minima**: a stationary point of E is generally not a stationary point of any E_n
5. It is less likely to stall near a saddle — some batch will have a real gradient there
6. There is evidence that the solutions it finds **generalize** better, which we return to in Class 18

Prince (2023), §6.2.2; Bishop & Bishop (2024), §7.2.3.

One honest caveat

- SGD does not converge in the classical sense: with a fixed η the iterate never stops moving
- The hope is that near a good solution every batch is well fitted, so every batch gradient is small
- In the next class we ask when that hope is justified, and add memory of past steps and a schedule for η

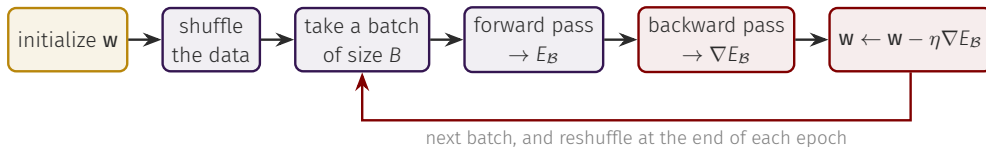
BATCH exact direction, expensive, destination fixed by the start

STOCHASTIC noisy direction, cheap, can change valley

This section rests on one substitution: replace ∇E by something with the *same mean* and a manageable variance.

Putting It Together

One training run, end to end



- Only the maroon boxes are new today, and only the last one is a choice

What we choose, and what chooses itself

Quantity	Set by	Governed by
∇E	the model and the data	backpropagation
L	the problem	nothing we control
η	us	must satisfy $\eta < 2/L$ (Lemma 2)
B	us	noise $\propto 1/\sqrt{B}$, cost $\propto B$ (Prop. 5)
shuffling	us	required for Proposition 5
$\mathbf{w}^{(0)}$	us	taken up in Class 15

Four of the six rows are ours to choose, and today gives a principled reason for three of them.

Takeaways

1. On a convex loss descent cannot fail: every stationary point is global, wherever we start
2. $-\nabla E$ is the steepest descent direction, and uniquely so (Lemma 1)
3. The eigenvalues of $\mathbf{H}$ are curvatures; $\lambda_{\max}$ is the sharpest one, and it sets L
4. A step with $\eta < 2/L$ is guaranteed to lower E ; a larger one can raise it (Lemma 2)
5. Gradients are worth their cost by a factor of W
6. Losing convexity costs the link between “stationary” and “best”; saddles, not local minima, are the practical difficulty
7. The mini-batch gradient is unbiased with standard error $\propto 1/\sqrt{B}$ (Prop. 5), and its noise is what stops the starting point being destiny

Next class: the step is safe, but how fast does it arrive?

References

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- H. Robbins & S. Monro, “A stochastic approximation method”, *Annals of Mathematical Statistics*, 1951.

Questions?